

EC3213 — Money, Credit and Banking

Winter 2023 Examination · Paper 1 · Full Marking Scheme

Module	EC3213 — Money, Credit and Banking
Internal Examiner	Ms. Aisling Finucane
Institution	University College Cork
Session	Winter 2023, Paper 1
Duration	1.5 hours
Structure	Section A — Long Questions: attempt 2 of 3 (100% of the paper). All questions carry equal weight.
Materials	Non-programmable calculator permitted

Coverage note. This scheme answers **all three** Section A questions in full, since you'll choose your best two under exam conditions. Note that A3 is worth the full 100 marks on its own — a single long essay question — so if you plan to answer A3, budget your 45-minute half of the exam accordingly.

Consistency with other sittings. A1(a) (fed funds rate/OMO) and A2(b) (money demand theories) draw on the same shared theory library as the Winter 2024–2025 EC3213 schemes. A2(a) here asks for *four* related concepts (traditional, expectations-augmented, and accelerationist Phillips curves, plus the Taylor rule) rather than the two or three asked on other sittings — this scheme distinguishes the expectations-augmented curve from the accelerationist curve explicitly, since conflating them is one of the most common ways marks are lost on this topic.

Source basis. A3's asset-price-bubble question follows the module's own "Lessons for Monetary Policy Strategy from the Global Financial Crisis" material almost directly. The broader academic debate referenced alongside it (Bernanke and Gertler's case against targeting asset prices, versus Cecchetti, Genberg and Wadhvani's case for leaning against credit-fuelled bubbles) reflects the

"independent research" the question explicitly invites — flagged as supplementary to, not a replacement for, the module's own material.

Section A · Question 1

The market for reserves, the discount rate/IOR corridor, and the money multiplier — 100 marks

A1(a)

[40 marks]

Explain, with the aid of a diagram, how the Federal Funds interest rate is affected by an Open Market Sale by the Federal Reserve Bank. Illustrate how Open Market Operations are an effective tool for targeting the Fed Funds Rate.

A1(b)

[40 marks]

Explain with the aid of a diagram how the Discount rate and the Interest on Reserve Balances minimise fluctuations in the Fed Funds rate due to changes in the demand for reserves.

A1(c)

[20 marks]

Derive and explain the money multiplier.

FULL MODEL ANSWER

The market for reserves: the federal funds rate is the price that clears the **market for bank reserves** — determined by the demand banks have for holding reserves (partly to meet requirements, partly as a cushion against unexpected deposit outflows) and the supply of reserves the Fed makes available.

The demand curve: downward-sloping in the fed funds rate — as i_{ff} falls, the opportunity cost of holding excess reserves (relative to lending them out) falls, so banks want to hold more of them. Since 2008 the Fed has paid interest on reserves (i_{or}) at a fixed spread below the fed funds target, so the demand curve **flattens out (becomes infinitely elastic) once i_{ff} falls to i_{or}** — below that point, banks would rather hold reserves than lend at a rate no better than what the Fed itself pays.

The supply curve: made up of non-borrowed reserves (supplied via OMOs) plus borrowed reserves (banks borrowing directly from the Fed at the discount rate, i_d). If $i_{ff} < i_d$, no bank will borrow from the Fed when cheaper funding

is available in the interbank market, so borrowed reserves are zero and supply is **vertical**. If i_{ff} rises above i_d , banks profitably borrow from the Fed and re-lend at i_{ff} , so supply becomes **horizontal (perfectly elastic) at i_d** .

The Market for Reserves: Effect of an Open Market Sale

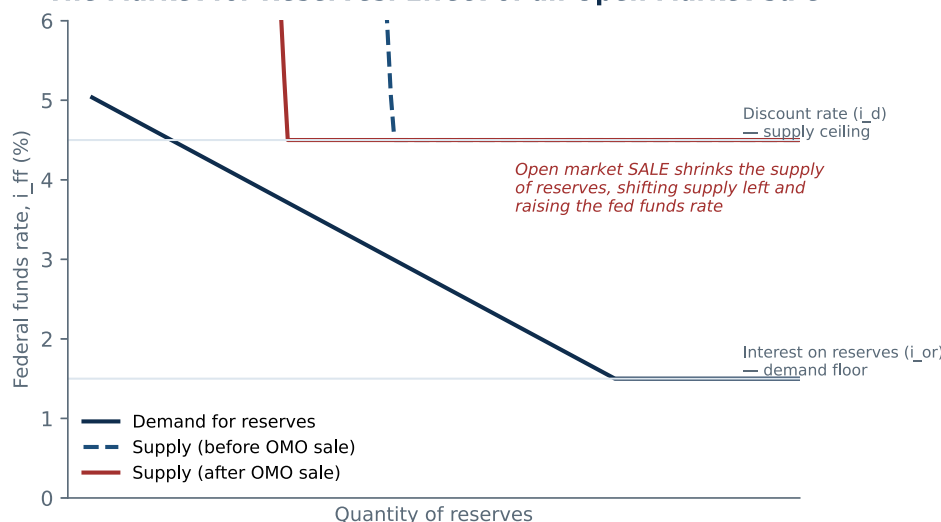


Fig. — An open market sale shrinks the supply of reserves, shifting the (vertical section of the) supply curve left. Where the new equilibrium remains on the downward-sloped section of the demand curve, this raises the federal funds rate from A to B.

Effect of an open market sale: an OMO sale drains reserves from the banking system, shifting the vertical portion of the supply curve to the **left**. Provided the new intersection with demand still falls on the downward-sloped section of the demand curve (i.e. the fed funds rate is still above i_{or}), this **raises the federal funds rate**. Conversely, an open market purchase shifts supply right and lowers i_{ff} .

Why OMOs are an effective targeting tool: OMOs give the Fed **complete control over the volume** transacted (unlike discount lending, where the bank itself decides whether to borrow), are **flexible and precise** — any desired quantity of reserves can be added or drained — and can be **quickly implemented and easily reversed** if conditions change, which is precisely what makes OMOs the Fed's primary conventional tool rather than the discount rate or reserve requirements.

EXAM TECHNIQUE

A common exam slip is describing the effect of an OMO sale without specifying *where* the new equilibrium falls relative to i_{or} — if reserves remain abundant enough that the intersection stays on the flat section of the demand curve at i_{or} , an OMO sale of typical size has **no effect** on the fed funds rate at all.

Mark Allocation — Part (a) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Demand and supply curves in the market for reserves correctly shaped (demand downward-sloping then flat at i_{or} ; supply vertical then flat at i_d) AND an OMO sale correctly shown shifting supply left, raising i_{ff} on the relevant section AND at least two specific reasons given for why OMOs are an effective targeting tool.
21-30	21-30	Correct diagram shape and correct direction of the OMO sale's effect, but the reasons OMOs are an effective tool are generic or missing.
11-20	11-20	Diagram shape roughly correct but the flat sections (at i_{or} and i_d) are missing or mislabelled, or the direction of the OMO sale's effect is reversed.

Framing — a different question from Part (a): Part (a) asked how a **deliberate policy action** (an OMO sale) moves i_{ff} by shifting supply. This part asks something different: how the fed funds rate stays **relatively stable** even when the demand for reserves shifts on its own — due to changing bank behaviour (e.g. heightened precautionary demand during a period of stress) — *without* any deliberate policy action at all.

The Discount Rate and IOR as a Corridor Bounding the Fed Funds Rate

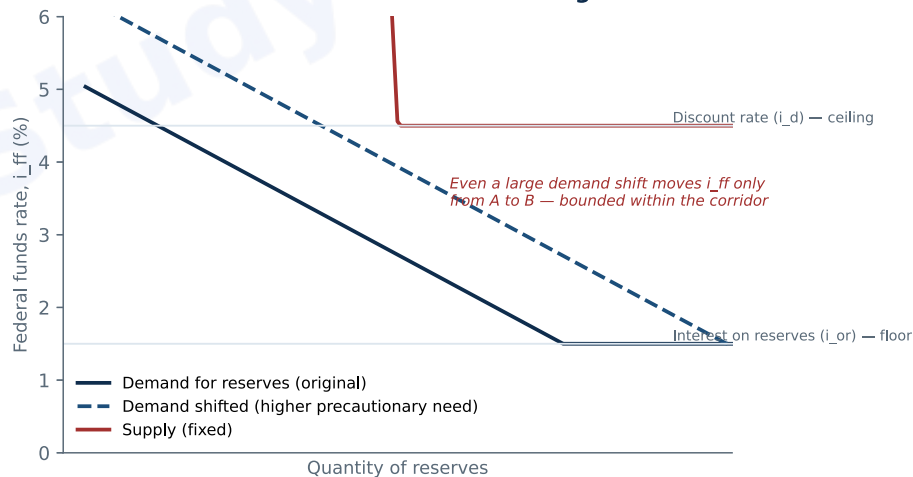


Fig. 1 — The discount rate (ceiling) and interest on reserves (floor) bound the fed funds rate within a corridor. A shift in the demand for reserves that would otherwise move i_{ff} substantially instead moves it only from A to B, since the curve flattens at each boundary.

How the interest on reserves floor works: if a shift in reserve demand (e.g. a fall in banks' desire to hold reserves) would otherwise push the equilibrium fed funds rate below i_{or} , it can't — because the demand curve itself goes flat at i_{or} (see Q. A1(a)): no bank will lend reserves overnight at a rate below what the Fed itself pays for holding them. The floor is therefore built directly into the shape of the demand curve.

How the discount rate ceiling works: if a shift in reserve demand (e.g. a rise in banks' precautionary demand for reserves, perhaps during a period of financial stress) would otherwise push the equilibrium fed funds rate above i_d , it can't — because the supply curve itself goes flat at i_d : once i_{ff} would exceed the discount rate, banks simply borrow directly from the Fed's discount window instead of bidding up the interbank rate further, capping i_{ff} at i_d .

The combined effect — minimising fluctuations: together, the two boundaries form a **corridor (or "channel") system**: as long as the fed funds rate target sits comfortably between i_{or} and i_d , demand-side shocks to the reserves market — of a kind the Fed did not initiate and may not even have anticipated — are automatically absorbed by banks either lending excess reserves at no less than i_{or} or borrowing shortfalls at no more than i_d , rather than being fully reflected in a volatile fed funds rate. This is precisely why the diagram in Fig. 1 shows a *large* shift in the demand curve producing only a *modest* change in the resulting fed funds rate (A to B) — the corridor absorbs most of the shock.

Mark Allocation — Part (b) (40 marks)

Tier	Marks	What separates this tier
31-40	31-40	Correctly distinguishes this from an OMO-driven shift (Part (a)) as a demand-side, not supply-side, disturbance AND correctly explains both

the i_{or} floor and i_{d} ceiling mechanisms **AND** the diagram correctly shows a demand shift bounded within the corridor, producing a smaller i_{ff} change than an unbounded shift would.

21-30	21-30	Both the floor and ceiling mechanisms correctly explained, but the diagram doesn't clearly show the demand shift being bounded, or the distinction from Part (a) isn't made.
11-20	11-20	Only one of the two boundaries (floor or ceiling) correctly explained.
0-10	0-10	No credible explanation of either boundary given.

Setting up the derivation: the monetary base MB is the sum of currency in circulation (C) and total reserves (R) held by banks: $MB = C + R$. The narrow money supply M1 is currency plus checkable deposits (D): $M1 = C + D$. Reserves themselves split into required reserves (a fraction, rr , of deposits) and any excess reserves banks choose to hold: $R = (rr \times D) + ER$.

Substituting through: $MB = C + (rr \times D) + ER$. Dividing the M1 definition by the MB definition, and dividing every term by D, lets us express both C and ER relative to deposits — define $c = C/D$ (the public's currency-to-deposit ratio) and $e = ER/D$ (banks' excess-reserve ratio). This gives:

Result: $m = M1/MB = (C+D)/(C + rr \cdot D + ER) = (c+1) / (c + rr + e)$

The simple special case: if we assume away currency holdings and excess reserves entirely ($c = 0$, $e = 0$) — i.e. all money stays in the banking system as deposits, and banks lend out every reserve dollar not legally required to be held — the formula collapses to the **simple money multiplier**: $m = 1/rr$. This is the same result reached by summing the geometric series of successive lending rounds directly.

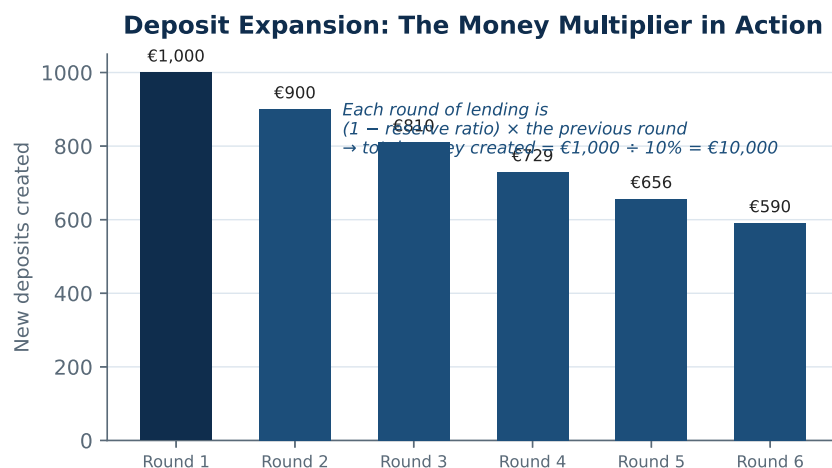


Fig. — The geometric-series intuition behind $m = 1/rr$: successive rounds of deposit expansion (illustrative: €1,000 initial deposit, 10% reserve ratio), summing to $€1,000 \div 10\% = €10,000$.

Round	New deposit created (€)	Held as reserves (€)	Re-lent (€)
1	1,000.00	100.00	900.00
2	900.00	90.00	810.00
...	(geometric series continues)		
Total money created	€10,000.00		= €1,000 ÷ 10%

Why the full derivation matters beyond the simple case: in reality c and e are both strictly positive — the public holds some currency outside banks, and banks hold some excess reserves as a cushion — so the **actual money multiplier $m = (c+1)/(c+rr+e)$ is always smaller than the simple $1/rr$ formula** would suggest. This matters directly for central bank policy: it's part of why the enormous expansion of the monetary base during quantitative easing (see the conventional/unconventional tools discussion) did not translate one-for-one into a proportional expansion of the broader money supply — c and e both rose sharply during and after the 2008 crisis as both the public and banks became more cautious.

Mark Allocation — Part (c) (20 marks)

Tier	Marks	What separates this tier
15-20	15-20	Correctly derives $m = (1+c)/(c+rr+e)$ from the MB and M1 balance-sheet definitions AND correctly shows the simple case $m = 1/rr$ AND gives a correct worked numeric example.
10-14	10-14	Correctly states the simple multiplier $m = 1/rr$ with a worked example, but the fuller algebraic derivation (via c and e) is missing or incomplete.
5-9	5-9	Vague reference to "banks lending out deposits" without a reserve ratio, multiplier formula, or derivation.
0-4	0-4	No credible derivation or explanation given.

Question A1 Total: 100 Marks

Section A · Question 2

The Phillips curve family, the Taylor Rule, and theories of money demand — 100 marks

A2(a)

[50 marks]

Outline the economic reasoning behind: (a) The Phillips Curve; (b) The Expectations Augmented Phillips Curve; (c) The Accelerationist Phillips Curve; and (d) The Taylor Rule.

A2(b)

[50 marks]

Compare the Classical Theory of Money Demand, the Keynesian Theory of Money Demand and Milton Friedman's Theory of Money Demand. In your opinion, which theory (theories) of money demand is most appropriate to describe the current economic environment? Why?

(a) The (traditional) Phillips Curve: A.W. Phillips's 1958 LSE study documented a **negative statistical relationship** between wage inflation and unemployment in the UK — tight labour markets were associated with higher wage inflation, and vice versa; Solow and Samuelson (1960) replicated the finding for the US using price inflation. The mechanism: lower unemployment increases workers' bargaining power, leading to higher nominal wage demands that feed through into higher prices — a wage-price spiral. Read as a stable, exploitable **policy trade-off**, this was the dominant view until Friedman's 1968 critique.

(b) The Expectations-Augmented Phillips Curve: Friedman's key insight was that workers bargain over **real**, not nominal, wages. In a tight labour market, workers would demand nominal wage growth in excess of whatever price inflation they *expect* — giving $\pi_t = \pi_t^e - \gamma(U_t - U^*)$, where expected inflation (π_t^e) now enters the relationship explicitly, but its own formation is left **unspecified** at this stage. This is a genuinely distinct, intermediate step from the accelerationist version below: it establishes *that* expectations matter and shift the curve, without yet committing to *how* those expectations are formed.

(c) The Accelerationist Phillips Curve: Friedman went further and reasoned specifically about how inflationary expectations are determined, arguing they are **adaptive** — this year's expected inflation is simply last year's actual inflation, $\pi_t^e = \pi_{t-1}$. Substituting this specific assumption into the expectations-augmented curve from (b) gives $\pi_t = \pi_{t-1} - \gamma(U_t - U^*)$: it is now the *change* in inflation, not its level, that relates to the unemployment gap. This means a central bank *can* hold unemployment below its natural rate U^* , but only by accepting **ever-accelerating** inflation — each period's attempt to exploit the trade-off pushes expected inflation up further, shifting the short-run curve outward again the following period.

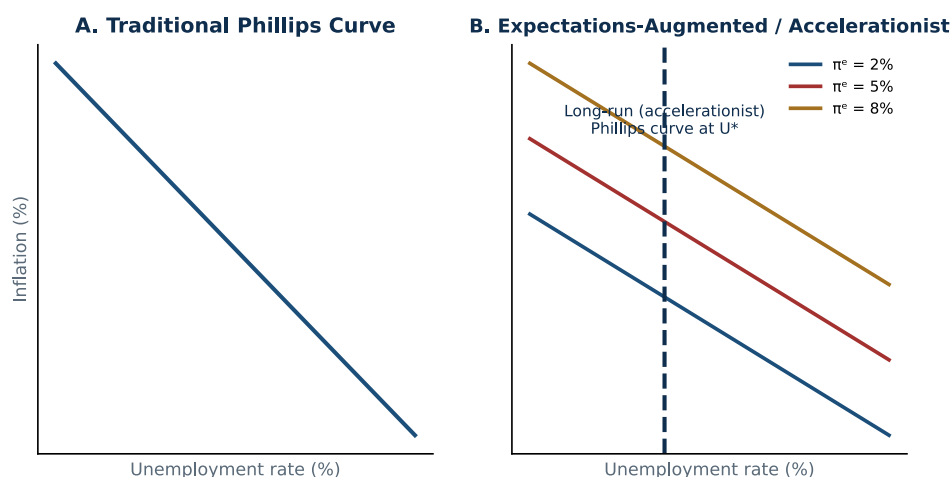


Fig. 1 — The traditional Phillips curve (A) suggests a stable trade-off; the expectations-augmented/accelerationist view (B) shows the curve shifting outward as expected inflation rises, with only a vertical long-run curve at the natural rate U^* .

Why the progression matters — and why the accelerationist model was vindicated: loose US fiscal and monetary policy in the 1960s initially produced exactly the traditional trade-off, but as the public's expectations adjusted, the trade-off worsened through the late 1960s, and by the 1970s produced **stagflation** — rising inflation *and* rising unemployment together, a positive correlation the original Phillips curve couldn't explain, but which the accelerationist model predicts exactly once expectations are allowed to adapt.

(d) The Taylor Rule: a simple, transparent rule prescribing how a central bank's policy rate should respond to observable economic conditions: $i_{ff} = \pi + r^* + \frac{1}{2}(\pi - \pi^*) + \frac{1}{2}(\text{output gap})$. Including *both* an inflation gap and an output gap reflects that a central bank cares not only about keeping inflation near target, but also about smoothing business-cycle fluctuations of output around its potential.

EXAM TECHNIQUE

The single most common way to lose marks on this question is treating (b) and (c) as the same thing. They are not: (b) establishes that expectations enter the Phillips curve at all (a general point, compatible with many different ways expectations might be formed); (c) commits to a *specific* mechanism — adaptive expectations — that turns the general expectations-augmented curve into the specific accelerationist prediction. Answering (c) correctly requires stating the adaptive-expectations assumption explicitly, not just repeating the expectations-augmented formula with a different name.

Mark Allocation — Part (a) (50 marks)

Tier	Marks	What separates this tier
38-50	38-50	All four correctly explained AND the expectations-augmented curve (b) is clearly distinguished from the accelerationist curve (c) via the explicit adaptive-expectations assumption AND the historical stagflation evidence is used to justify the progression from (a) to (c).
25-37	25-37	All four named with broadly correct formulas, but (b) and (c) are conflated or not clearly distinguished.
13-24	13-24	Only two or three of the four correctly explained.
0-12	0-12	Only one, or none, correctly explained.

Classical Theory of Money Demand (Fisher's Quantity Theory): built from the equation of exchange, $MV = PY$. Fisher assumed **velocity (V) is fairly constant** in the short run (a matter of fixed institutional payment habits) and that output (Y) sits at its full-employment level given flexible wages and prices. Money demand is then $M_d = (1/V)PY = kPY$ — a fixed fraction of nominal income, with **no role for interest rates** at all. This makes the theory a clean theory of inflation: with V and Y essentially fixed, $\% \Delta M \approx \% \Delta P$.

Keynesian Liquidity Preference Theory: Keynes identified three motives for holding money — **transactions** and **precautionary** motives (both roughly proportional to income, as in the classical view) and, crucially, a **speculative** motive: money held as a store of wealth, as an alternative to holding bonds, when expected bond returns look unattractive. This makes money demand $M_d/P = f(i, Y)$ — negatively related to the interest rate as well as positively related to income — which means **velocity is not constant**: it moves procyclically with interest rates and with changing expectations about future "normal" interest rate levels, undermining the classical link between the money supply and nominal spending.

Friedman's Modern Quantity Theory: Friedman treated money as just another asset in a portfolio-choice framework, with demand for *real* money balances driven by **permanent income** (a longer-run, smoothed measure of wealth, rather than current income) and the expected returns on the alternative assets households could hold instead — bonds, equities, and physical goods (via expected inflation) — each relative to the return on money itself. Friedman argued that because these assets are not close substitutes for money in most portfolios, money demand is only **weakly sensitive to interest rates** — closer to the classical view in its practical stability, but

grounded in explicit portfolio-choice reasoning rather than a fixed institutional velocity assumption. This is why Friedman's theory implies velocity is *not* literally constant (as Fisher assumed) but is **stable and predictable** — a crucial practical distinction from the Keynesian view, where velocity's sensitivity to volatile interest-rate expectations makes it genuinely unpredictable.

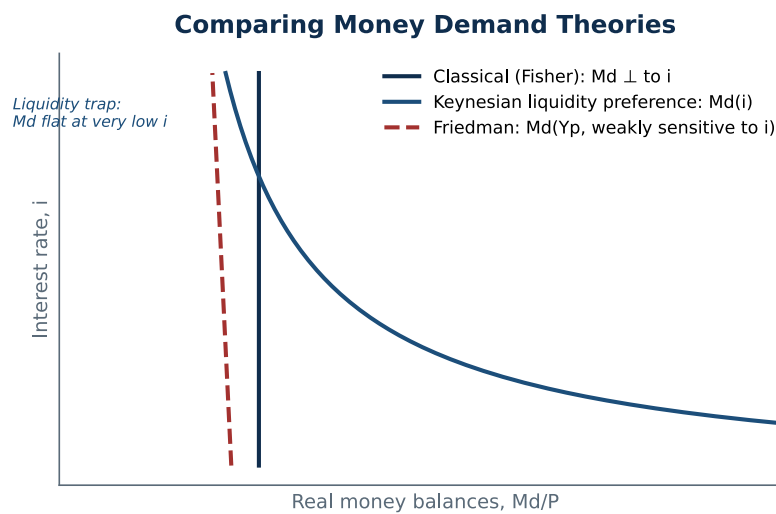


Fig. — Comparing the three theories: Classical money demand is vertical (interest-rate-independent); Keynesian liquidity preference slopes down, including a flat "liquidity trap" region at very low rates; Friedman's is only weakly interest-sensitive, positioned by permanent income rather than current income.

	Classical (Fisher)	Keynesian (liquidity preference)	Friedman (modern quantity theory)
Interest-rate sensitivity	None	Strong (negative)	Weak
Income measure	Current nominal income	Current income	Permanent income
Velocity	Constant	Unstable/unpredictable	Stable, though not constant
Policy implication	Money supply tightly linked to nominal spending	Money supply loosely, unreliably linked to spending	Money supply reliably, though not mechanically, linked to spending

Opinion: which theory best fits the current economic environment?

Assessment: the most defensible position is that **Friedman's modern quantity theory offers the more useful practical framework today**, but not because interest rates are irrelevant — rather because the Keynesian liquidity preference framework's core policy implication (that velocity is so unstable the money supply is barely worth tracking) is less useful in an environment where most major central banks explicitly target an **interest rate** (not a monetary aggregate) as their primary instrument. In that sense, elements of *both* Keynesian and Friedman's reasoning show up in how central banks actually operate: interest-rate targeting (the Keynesian-consistent practical response to unstable velocity) implemented with an eye toward the same permanent-income-driven, portfolio-based stability Friedman emphasised, rather than the classical view's simple, mechanical $MV=PY$ link, which fits poorly with a world of quantitative easing where enormous expansions of the monetary base did not translate one-for-one into proportional inflation (see the money-multiplier derivation in Q. A1(c), where

the same widening gap between the simple and actual multiplier tells the same story from the supply side).

EXAM TECHNIQUE

This is a genuinely open question — a well-reasoned case for the Keynesian view or the classical view can score equally well, provided the argument connects the theory's own assumptions to specific, observable current conditions rather than simply asserting a preference.

Mark Allocation — Part (b) (50 marks)

Tier	Marks	What separates this tier
38-50	38-50	All three theories correctly explained with their core mechanisms AND a clear opinion given AND the opinion is justified by connecting the theory's own assumptions to specific, observable current conditions.
25-37	25-37	All three theories correctly explained, and an opinion given, but the justification is generic or doesn't engage with current conditions specifically.
13-24	13-24	Only two of the three theories correctly explained, or all three explained with no opinion/justification given at all.
0-12	0-12	Fewer than two theories correctly explained.

Question A2 Total: 100 Marks

Section A · Question 3

Should monetary policy react to asset prices and asset bubbles? — 100 marks

A3(a)

[100 marks]

"Monetary policy should react to asset prices and asset bubbles." Based on material covered in class and your independent research, discuss and critique the above statement. You may also discuss central bank regulatory policy to support your answer.

FULL MODEL ANSWER

EXAM TECHNIQUE

This is a single, 100-mark essay question — the full length of your answer for this question, rather than a short-question-style response. The structure below moves from definitions, through the module's own "lessons from the financial crisis" material, to the wider academic debate, and finishes with a clear, defended verdict — that overall shape (define → apply the class framework → engage the debate →

What an asset-price bubble is, and why the statement needs unpacking before it can be answered

Definition: an asset-price bubble is a pronounced increase in asset prices that departs from the assets' underlying fundamental values, and which eventually bursts. Not all bubbles are alike, and the module's own material distinguishes two types that turn out to matter enormously for whether the statement in the question is true.

- **Bubbles driven solely by irrational exuberance:** asset prices detach from fundamentals through speculative enthusiasm alone, without a corresponding surge in credit — e.g. a bubble in an asset class purchased mostly with investors' own capital.
- **Credit-driven bubbles:** asset prices and the volume of credit extended to fund their purchase rise rapidly *together* — the defining feature of the run-up to the 2007–2008 subprime financial crisis, where mortgage credit and house prices rose in lockstep.

Why this distinction is the crux of the question: the statement "monetary policy should react to asset prices and asset bubbles" is stated as a single, unconditional claim — but the module's own "lessons from the financial crisis" material and the wider academic debate both converge on the conclusion that **the right answer depends entirely on which type of bubble is being discussed**, which is exactly why the statement, taken at face value, is too broad to simply agree or disagree with.

The case against monetary policy reacting to bubbles generally

The traditional ("Greenspan doctrine") view: there is a strong argument for **not** having monetary policy react to bubbles driven by irrational exuberance specifically — the prevailing pre-crisis approach was that central banks should not try to prick bubbles pre-emptively, and should instead "mop up" afterward, easing policy once a bubble bursts rather than raising rates to deflate it in advance. The reasoning: a central bank cannot reliably tell in real time whether an asset price increase is a bubble or a justified re-pricing of fundamentals; and the policy rate is a **blunt instrument** — raising it enough to meaningfully deflate a bubble concentrated in one asset class (e.g. tech stocks, or a specific property segment) would simultaneously depress borrowing, investment and employment across the *entire* economy, a very high cost to pay for addressing a problem in one corner of it.

The academic version of this case — Bernanke and Gertler: Bernanke and Gertler (1999, 2001) formalise this into their well-known prescription within an inflation-targeting framework: **asset prices should affect monetary policy only to the extent that they affect the central bank's forecast of inflation** — not as a separate target in their own right. Their argument is that a policy rule reacting directly and aggressively to asset prices performs no better, and can perform worse, than one that reacts to asset prices only indirectly, through their effect on the inflation outlook.

The case for monetary policy (or, more precisely, some form of policy) reacting to credit-driven bubbles specifically

Why credit-driven bubbles are different: the module's own "lessons from the financial crisis" material is explicit that **credit-driven bubbles are easier to identify** than pure irrational-exuberance bubbles, precisely because they carry a second, independent, corroborating signal: asset prices and the volume

of credit are both increasing rapidly at the same time. Where a irrational-exuberance bubble has to be inferred from price behaviour alone (inherently hard to distinguish from a genuine fundamentals-driven boom), a credit-driven bubble shows up in *two* independent data series moving together, which is a materially more reliable signal.

The academic case for a more activist response — Cecchetti, Genberg, Lipsky and Wadhvani: this group's Geneva Reports (2000, 2002) directly challenge the Bernanke-Gertler prescription, arguing that policymakers who set rates based purely on a fixed-horizon inflation forecast could do better by **reacting directly to asset-price misalignments** — "leaning against the wind" of a developing credit-and-asset-price boom rather than waiting to mop up after it bursts — potentially producing a smoother path for both output and inflation than a purely inflation-forecast-based rule.

Why credit-driven bubbles are also more *dangerous*, not just more identifiable: when a credit-driven bubble bursts, the resulting deleveraging — borrowers and the banks that lent to them all simultaneously trying to repair balance sheets — amplifies the downturn through the financial system itself, which is exactly the mechanism behind the severity of the 2007–2008 crisis and why the module's material concludes that regulators **should not have a laissez-faire attitude and let credit-driven bubbles proceed without any reaction** at all — a direct, explicit rejection of extending the "don't react to bubbles" doctrine to this specific case.

The critique and resolution: the right tool is not necessarily the policy rate

Reconciling the two positions — the statement is broadly right about needing to act, but wrong about the tool: the strongest, most defensible position — and the one the module's own material points toward — is that the blanket statement "monetary policy should react to asset prices and asset bubbles" is **overbroad as written**. It should be rejected as a general rule for irrational-exuberance bubbles (the Bernanke-Gertler case remains persuasive there), but a *version* of it is right for credit-driven bubbles specifically — except that the appropriate tool is better understood as **macroprudential regulation** rather than the blunt instrument of the policy interest rate.

What macroprudential policy offers instead: regulatory policy aimed directly at credit-market conditions in aggregate — countercyclical capital buffers, loan-to-value and debt-to-income limits on mortgage lending, and stress testing — can target the specific sector where a credit-driven bubble is forming without raising the cost of borrowing for the entire economy the way a policy-rate increase would. This directly addresses the core objection to using monetary policy against bubbles (its bluntness) while still delivering the "don't just wait and mop up" response the module's material calls for in the credit-driven case specifically — the same countercyclical capital buffer discussed in the EC3005 Winter 2022 scheme, Q. A1(c), is a direct example of this kind of targeted tool.

Verdict

Disagree with the statement as a general rule for monetary policy; agree that credit-driven bubbles warrant an active response — through macroprudential regulation, not primarily the policy interest rate.

Summary of the argument: the case against reacting to bubbles generally (identification difficulty, the bluntness of the policy rate, Bernanke-Gertler's formal case) is strong and should not be dismissed. But it applies most cleanly

to irrational-exuberance bubbles; credit-driven bubbles are both more identifiable (the dual price-and-credit signal) and more dangerous (financial-system amplification on the way down) in a way that justifies a genuinely different policy conclusion — not necessarily reacting with the policy interest rate, which remains a blunt instrument even for credit-driven bubbles, but through macroprudential regulatory tools that can target the specific credit build-up directly. A student arguing the opposite conclusion (that monetary policy, via the interest rate, should react broadly) can score equally well provided the argument directly engages with, rather than ignores, the identification-difficulty and bluntness objections raised by the Bernanke-Gertler side of the debate.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
76-100	76-100	Correctly defines asset-price bubbles and distinguishes credit-driven from irrational-exuberance types AND presents both sides of the academic debate (Bernanke-Gertler vs. Cecchetti et al.) accurately AND engages with the module's own "lessons from the financial crisis" material specifically AND reaches a clear, well-defended verdict that distinguishes the appropriate tool (monetary policy vs. macroprudential regulation) for different bubble types.
51-75	51-75	Correctly defines bubble types and presents a reasonable case for at least one side of the debate, with a verdict given, but doesn't fully engage with the opposing view or doesn't distinguish which policy tool (rates vs. regulation) is appropriate for which bubble type.
26-50	26-50	Discusses asset price bubbles in general terms without the credit-driven/irrational-exuberance distinction, or without engaging with the named academic debate, or without a clear verdict.
0-25	0-25	No credible discussion of the statement given, or the answer doesn't engage with monetary policy or asset price bubbles specifically.

Question A3 Total: 100 Marks